V2X-GW

PRODUCT REQUIREMENTS DOCUMENT (PRD)

Low-Latency V2X Edge Gateway

Internet of Things / Smart Infrastructure | Rev A1 - Architecture, Atomic Requirements, Interfaces & Verification Baseline

Mission profile. The gateway supports 5G/C-V2X communication between vehicles and smart infrastructure with a sub-20 ms mission-level latency target. That target is meaningful only after the owned measurement boundary, message class, timestamp point, traffic load, radio condition and infrastructure egress are defined. The product therefore treats latency as a controlled end-to-end budget rather than a single processor benchmark, and it degrades safely when time synchronization or data validity can no longer support trustworthy forwarding.

Engineering baseline. The released architecture uses an NXP i.MX 8 family compute platform with C-V2X baseband capability, a qualified RF front end and antenna system, GNSS/time synchronization, Yocto Linux with containerized V2X messaging services, PoE or 12/24 V DC input, passive cooling, SCMS PKI anchored in protected hardware, and a NEMA 4X roadside enclosure. BSM/SPaT message handling, secure boot/update, RF tuning, timing validation and certificate provisioning are configuration controlled; missing qualification or production evidence blocks release rather than being represented as a fictitious runtime state.

Definition	Controlled product interpretation
Product class	Low-latency 5G/C-V2X edge gateway for smart-city infrastructure and autonomous-fleet deployments.
Safety / assurance	Fail-safe degradation when C-V2X timing synchronization or message validity is lost; the gateway does not assume vehicle-motion authority.
Lifecycle mission	15-year lifecycle intent with remote telemetry of RF degradation, credential maintenance, controlled software updates and field-service traceability.
Primary architecture	NXP i.MX 8 family processor/baseband, RF front end and antennas, GNSS/time synchronization, Yocto Linux containers, SCMS hardware trust anchor, PoE or 12/24 V input, passive cooling and NEMA 4X packaging.
Release philosophy	Runtime faults use bounded degraded behavior; incomplete RF, timing, cybersecurity, environmental, manufacturing or supplier evidence blocks the affected variant from production equivalence.

Future radio modules, antenna variants, regional V2X profiles, roadside installations and SCMS backends are supported only through explicit interface parameters, controlled configuration identities and targeted regression when assumptions change.

PRODUCT GOVERNANCE, REQUIREMENT QUALITY & LATENCY OWNERSHIP

Product requirements separate product-owned behavior from installation assumptions, radio-network variability, regional message profiles and supplier-dependent characteristics. The sub-20 ms mission target is maintained as a controlled budget with named ingress and egress timestamps, message class, traffic condition and test configuration. No latency claim may be generalized from an unloaded processor measurement or a single RF condition. Security services, diagnostics and maintenance functions are included in the same timing and resource envelope when they can affect forwarding.

Common release evidence includes hardware revision, software/container image, calibration and RF configuration, trust-anchor and credential state, antenna/cable variant, test instrumentation, environmental condition and network load. Values not fixed by the program remain TBC with an owner and closure method. Runtime degradation is reserved for detected faults or invalid data; gaps in qualification, DVP, manufacturing capability or supplier equivalence keep the release gate open.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-GOV-001	Each product requirement shall trace to a mission need, risk control, interface obligation or lifecycle objective and to objective verification evidence.	Bidirectional trace and DVP coverage audit.
V2X-GOV-002	The sub-20 ms latency target shall be configuration controlled by defined measurement points, message class, traffic load and operating condition.	Latency-budget review plus synchronized timestamp correlation.
V2X-GOV-003	Unclosed qualification, manufacturing, cybersecurity or supplier evidence shall block the affected configuration from release.	Release-gate and configuration-status audit.

Verification, margin and future-change rule

A change in owned latency boundary, radio module, operating system, container topology, credential model, antenna system or installation environment reopens only the analyses and tests whose assumptions are invalidated. Generic planning values shall not be promoted into released limits without owner-approved evidence.

DEPLOYMENT BOUNDARY, DATA AUTHORITY & EXPLICIT EXCLUSIONS

The gateway sits between radio-facing V2X functions, local roadside or fleet infrastructure networks and management services. It may receive vehicle-originated messages, infrastructure state such as SPaT, GNSS/time information, local diagnostics and credential status. It does not create vehicle-motion authority, override local traffic-signal safety interlocks or treat cloud reachability as a prerequisite for immediate data-validity decisions. Every interface that can influence message forwarding must define ownership, freshness, integrity, timeout and fallback behavior.

The product boundary also identifies installation-owned items such as antenna location, mast or cabinet grounding, external surge protection, Ethernet plant, PoE source capability, local power feeder, GNSS sky view and backhaul quality. Those dependencies are captured in the installation ICD and may not be silently credited as product-internal safety or performance mechanisms. A failed external dependency results in a bounded product state and explicit status rather than hidden substitution of stale data.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-SYS-001	Each external data interface shall define source, destination, encoding, freshness, integrity, timeout and non-permissive fallback.	ICD review with stale/range/integrity fault injection.
V2X-SYS-002	The gateway shall not assert vehicle-motion or traffic-signal safety authority that is not explicitly allocated to the product.	System boundary and authority review.
V2X-SYS-003	Installation dependencies credited for performance or protection shall be identified and verified at the deployed interface.	Installation acceptance and dependency audit.

Verification, margin and future-change rule

Changes to deployment topology, backhaul ownership, local traffic-controller interface or radio-service allocation require an interface impact review. External mitigations may be reused only when their timing, availability and independence remain bounded.

END-TO-END LATENCY BUDGET & TIMESTAMP ARCHITECTURE

Low latency is treated as a chain of measurable contributors: radio reception, baseband handoff, kernel and driver scheduling, container processing, message validation, application transformation, queueing, network transmission and egress timestamping. The released budget identifies where each timestamp is generated and which clock domain it uses. Queue depth, diagnostic load, certificate operations and concurrent network traffic are included because nominal one-message measurements do not represent a congested roadside node.

The gateway distinguishes processing latency from over-the-air propagation and from infrastructure-network delay outside the owned boundary. Message classes with different urgency may use different budget allocations, but none may silently bypass validity or security checks to recover timing. When the synchronized clock is invalid, the product must stop making precise age or latency claims and transition affected services to a defined degraded behavior.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-TIM-001	The released latency budget shall allocate maximum permitted contribution to each owned processing and queueing stage.	Instrumented end-to-end timestamp trace under representative load.
V2X-TIM-002	All latency measurements used for release shall identify clock source, synchronization state and timestamp insertion point.	Timing architecture review and correlation test.
V2X-TIM-003	A loss of qualified time shall invalidate dependent latency and message-age calculations before they can increase forwarding permissiveness.	Time-loss fault injection with output/status capture.

Verification, margin and future-change rule

Processor, kernel, driver, container, network-stack or time-source changes reopen the affected latency contributors rather than the entire product by default. Budget margin is retained as an owned parameter and verified under the maximum defined concurrent load.

SYSTEM ARCHITECTURE & RADIO-TO-EDGE DATA FLOW

The architecture decomposes the gateway into power entry, compute/baseband, RF front end, antenna paths, GNSS/time services, local Ethernet or PoE, protected credential storage and management interfaces. Data flow diagrams show where messages cross trust boundaries and where validity, freshness and authorization are checked. The preferred default is fail-silent for a service whose required input or timing basis is invalid, while independent management and diagnostic channels remain available only within their authorized roles.

Reset domains and startup order are explicit. Power good, secure boot, trusted software image, time-source qualification, radio readiness, credential readiness and application health are established before normal forwarding is advertised. A partial restart of one container or radio function must not cause stale queues from an earlier epoch to be replayed as current information.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-ARC-001	The released block diagram shall identify power, radio, compute, time, credential and network trust boundaries.	Architecture and interface review.
V2X-ARC-002	Normal forwarding shall become permissive only after required power, software, time, radio and credential states are qualified.	Startup-state and partial-restart test.
V2X-ARC-003	Reset or restart of one service shall not replay stale queued data as current after the service returns.	Service-restart and queue-epoch fault test.

Verification, margin and future-change rule

Topology changes such as split radio/compute modules, additional network interfaces or a different trust-anchor location require renewed boundary and startup-state analysis, with regression limited to the modified data and authority paths.

POE / 12 V / 24 V POWER ENTRY, PROTECTION & STARTUP

The product supports PoE or 12/24 V DC deployment variants. Each variant defines its own nominal range, transient envelope, source impedance, inrush behavior, reverse-polarity or miswire protection where applicable, surge suppression, conducted-EMI filtering and startup sequencing. Power architecture prevents an input disturbance from producing uncontrolled radio transmission, corrupted credential storage or repeated reset behavior that leaves external systems with ambiguous gateway status.

Logic rails, RF rails and any isolated Ethernet power domains are supervised so brownout is detected before data integrity is compromised. Retained configuration is validated after reset rather than assumed correct. If the installation supplies external surge or lightning protection, its location and residual waveform are interface parameters and must be verified in the deployed configuration.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-PWR-001	Each PoE or DC-input variant shall have a released electrical envelope, protection concept and startup/shutdown state sequence.	Power-tree review plus input-corner and transient test.
V2X-PWR-002	Brownout shall force bounded reset or inhibit behavior before software, radio or credential state can become indeterminate.	Brownout ramp and repeated-reset test.
V2X-PWR-003	External surge protection credited by the design shall be an explicit installation dependency with verified residual stress.	Surge coordination analysis and installation test.

Verification, margin and future-change rule

Changes to PoE PD implementation, DC/DC regulators, surge components, feeder length or cabinet grounding reopen input stress, reset and conducted-EMI verification. Production substitutions are not accepted from pin compatibility alone.

NXP i.MX 8 FAMILY COMPUTE / BASEBAND PLATFORM

The compute platform provides application processing, network handling and interfaces to the C-V2X radio/baseband implementation. CPU cores, memory, storage, accelerators and I/O are budgeted against the maximum defined message, diagnostic, security and management load. A family designation does not make all devices interchangeable; exact silicon variant, memory population, BSP, silicon errata disposition and clock/power configuration are controlled by the released hardware/software variant.

Health monitoring detects processor reset causes, watchdog events, memory faults and persistent resource exhaustion. Critical message paths are designed so a nonessential container or telemetry function cannot consume all CPU, memory, storage I/O or network descriptors needed for time-bounded V2X processing. Where hardware isolation features are used, their configuration becomes part of the release evidence.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-HW-001	The released compute variant shall define CPU, memory, storage, I/O, clock and silicon-errata assumptions used by performance analysis.	BOM/BSP/configuration audit.
V2X-HW-002	Critical V2X processing shall retain measured resource reserve under the maximum defined concurrent workload.	Stress test with CPU, memory and I/O telemetry.
V2X-HW-003	A nonessential workload shall not exhaust shared resources required for time-bounded forwarding and validity checks.	Resource-starvation fault injection.

Verification, margin and future-change rule

Changing the i.MX 8 variant, memory device, BSP, storage technology or clock policy requires targeted performance, timing, thermal and fault-recovery regression. Benchmark equivalence without shared-resource evidence is insufficient.

C-V2X BASEBAND & RF FRONT-END ARCHITECTURE

The radio path includes baseband processing, RF transceiver functions, power amplification where applicable, low-noise receive path, filtering, switching, duplexing or coexistence elements and antenna feed. The design controls conducted power, receive sensitivity, linearity, adjacent-channel behavior, spurious emissions, temperature drift and protection from installation-induced mismatch within the project-selected radio profile. Values not defined by the portfolio remain controlled parameters rather than invented universal limits.

RF enable is coordinated with oscillator lock, time validity, calibration state, thermal state and regulatory or installation permissions. Faulted RF hardware shall not be compensated by unconstrained software gain or power commands. Diagnostic telemetry distinguishes unavailable radio, degraded link quality and external interference so higher layers do not treat all reception loss as the same root cause.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-RF-001	The released RF chain shall configuration-control active devices, filters, switches, matching, calibration data and antenna-port assumptions.	RF schematic/BOM/calibration review.
V2X-RF-002	Transmit and receive performance shall remain within released limits across supply, temperature, channel and antenna-mismatch conditions.	Conducted and radiated RF characterization.
V2X-RF-003	RF faults or invalid calibration shall not be masked by software commands that exceed approved power or gain bounds.	Calibration corruption and RF-fault injection.

Verification, margin and future-change rule

Radio-module, PA/LNA, filter, oscillator, matching-network or PCB-stack changes reopen only the affected RF, thermal, EMC and calibration evidence, including coexistence with GNSS and local digital interfaces.

GNSS / TIME SYNCHRONIZATION ARCHITECTURE

Time synchronization underpins message age, event ordering and any latency assertion. The gateway combines GNSS or project-approved timing sources with local oscillator holdover and software time services. It tracks synchronization quality, source identity, offset, drift and transitions between qualified and degraded states. The design does not equate a running clock with a trusted clock; it requires evidence that the source and current uncertainty remain within the released application envelope.

Loss of GNSS view, antenna fault, time-source disagreement or excessive drift is handled without silently continuing precise timestamp claims. Services whose correctness depends on synchronized time suppress or mark affected outputs according to the released interface contract, while functions that do not require absolute time may continue if their own prerequisites remain valid.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-TIM-004	The gateway shall maintain an explicit time-quality state including source, validity, offset or uncertainty and holdover status.	Time-service trace under acquisition, loss and recovery.
V2X-TIM-005	Time-dependent forwarding shall use only timestamps whose quality state meets the released message contract.	Boundary and stale-time injection test.
V2X-TIM-006	Loss or disagreement of time sources shall produce bounded degraded behavior without preserving a false synchronized indication.	GNSS disconnect/source-disagreement test.

Verification, margin and future-change rule

GNSS receiver, antenna, oscillator, time daemon or synchronization algorithm changes require renewed acquisition, drift, holdover and recovery characterization, including temperature and power-cycle corners.

ANTENNA SYSTEM, FEEDLINES & INSTALLATION CONTROL

The antenna architecture is part of the communication system rather than a passive accessory. The released installation controls antenna type, polarization, placement, separation from GNSS and other radios, feedline type and length, connector family, grounding/bonding and any diversity arrangement. The NEMA 4X enclosure and antenna feedthroughs must preserve RF performance while meeting environmental sealing and field-service constraints.

Mismatch, cable loss, water ingress, connector corrosion and mounting changes can consume link margin without producing an obvious digital fault. The gateway therefore records useful RF-health indicators and supports installation acceptance checks that can distinguish degraded antenna path from compute or network faults. Any auto-tuning mechanism is bounded by released calibration limits.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-RF-004	Antenna type, placement, feedline, connector, grounding and diversity assumptions shall be released as installation-controlled parameters.	Installation ICD and antenna-system review.
V2X-RF-005	The antenna path shall retain released performance after environmental exposure and representative cable/connector tolerances.	Radiated/VSWR/cable-loss correlation before and after environment test.
V2X-RF-006	Automated tuning shall remain inside approved matching and transmit-stress limits and shall report failure to converge.	Tuning-range and mismatch fault test.

Verification, margin and future-change rule

Antenna, cable, feedthrough, enclosure location or roadside mounting changes trigger installation-specific RF and environmental correlation rather than a generic software-only release review.

ETHERNET / POE BACKHAUL & LOCAL NETWORK INTERFACES

Ethernet may carry PoE power, management traffic, infrastructure messages and backhaul data. The design separates timing-critical V2X traffic from lower-priority telemetry and maintenance where they share physical resources. Link state, negotiated mode, queue occupancy, packet errors and network congestion are observable so degraded external networking is not misclassified as radio failure. The product defines behavior for loss of backhaul without assuming cloud availability for local validity decisions.

Network interfaces are protected from malformed or unauthorized traffic through project-approved filtering and authentication controls. Maintenance access cannot bypass local message-age, time-quality or resource limits. Where multiple VLANs, ports or network zones are used, their separation and routing authority are configuration controlled and included in overload testing.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-NET-001	Network interfaces shall define traffic classes, authority, priority and resource limits for V2X, management and telemetry flows.	Network architecture and QoS review.
V2X-NET-002	Backhaul loss or congestion shall not cause stale infrastructure data to be forwarded as current.	Congestion and link-loss fault injection.
V2X-NET-003	Unauthorized maintenance or network traffic shall not bypass local timing, validity or resource-protection rules.	Negative authorization and flood testing.

Verification, margin and future-change rule

Switch, PHY, PoE magnetics, VLAN policy, network topology or backhaul changes reopen network timing, overload, EMC and cybersecurity evidence for the affected interfaces.

YOCTO LINUX BASE PLATFORM & BSP CONTROL

The software baseline uses Yocto Linux as a controlled platform rather than an open-ended distribution. Kernel version, BSP, drivers, device tree, security configuration, filesystem policy, service set, time synchronization stack and package manifest are frozen for each release. The build is reproducible and traceable to the hardware variant so field issues can be correlated to the exact kernel and driver state.

Startup order establishes secure boot status, persistent-storage integrity, time service, radio interfaces, network interfaces and application containers in a defined sequence. Unneeded services are disabled, watchdog and logging behavior are bounded, and diagnostic collection cannot exhaust storage or I/O needed by V2X applications. Recovery from a failed service is tested under live message load.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-SW-001	Each released software image shall identify Yocto manifest, kernel/BSP, drivers, configuration and hardware compatibility.	Reproducible-build and software-bill audit.
V2X-SW-002	Startup dependencies shall prevent applications from advertising normal operation before required platform services are valid.	Cold/warm boot sequencing test.
V2X-SW-003	Logging or diagnostic activity shall not exhaust storage or I/O required by time-bounded V2X processing.	Storage-fill and diagnostic-stress test.

Verification, margin and future-change rule

Kernel, BSP, driver, package or filesystem-policy changes require impact analysis against timing, security, startup, resource and RF-interface assumptions; a package update is not treated as low risk merely because application APIs remain unchanged.

CONTAINERIZED V2X SERVICES & RESOURCE PARTITIONING

Containerization separates messaging stacks and support services operationally, but it is not credited as physical independence without resource evidence. CPU quotas, memory limits, network namespaces, file access, device access, restart policy and inter-container interfaces are released configuration. Critical message services retain capacity when telemetry, diagnostics or management containers are intentionally overloaded.

Container lifecycle management clears stale queues and verifies configuration before a restarted instance re-enters service. Shared libraries, kernel facilities, storage and network drivers remain common-cause resources and are included in failure analysis. An orchestrator or supervisor may restart a failed container, but it cannot widen message authority or freshness windows as part of recovery.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-SW-004	Container CPU, memory, device, network and storage permissions shall be least-privilege and configuration controlled.	Container manifest and runtime-policy audit.
V2X-SW-005	A failed or overloaded noncritical container shall not consume the reserved resources of critical V2X message services.	Cross-container stress and starvation test.
V2X-SW-006	Container restart shall invalidate stale queues and re-establish prerequisites before outputs are declared valid.	Crash/restart test with message-epoch tracing.

Verification, margin and future-change rule

Changes to runtime engine, namespace model, shared storage, orchestration or container boundaries reopen the affected FFI, timing, restart and cybersecurity evidence. Software consolidation is assessed for common-cause impact.

BSM / SPAT MESSAGE INGEST, VALIDATION & OUTPUT CONTRACTS

The gateway supports containerized V2X message stacks including BSM and SPaT classes. For each released profile, the data contract defines message version, mandatory fields, units or encoding, producer identity, timestamp semantics, freshness, plausibility, duplicate handling and behavior on missing fields. The gateway does not silently normalize an invalid field into a plausible value when doing so could hide the loss of trustworthiness.

Transformation between radio and infrastructure representations is deterministic and testable. Unknown extensions are handled according to the profile policy, and schema or version changes cannot become active without compatibility evidence. Message output includes sufficient status for the receiving system to know whether data is current, qualified, degraded or rejected.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-MSG-001	Each supported message profile shall define schema/version, field validity, freshness and fallback behavior.	Protocol/profile conformance review.
V2X-MSG-002	Invalid or stale safety-relevant fields shall not be converted into apparently valid current data.	Malformed/stale message negative test.
V2X-MSG-003	Message transformation shall be deterministic and traceable from received payload to emitted representation.	Golden-vector and round-trip correlation test.

Verification, margin and future-change rule

BSM/SPaT profile, schema, encoding or regional deployment changes require targeted protocol, timing and compatibility regression. New optional fields are not automatically accepted as trusted inputs to existing application logic.

EDGE PROCESSING PIPELINE, FILTERING & DETERMINISTIC OUTPUT

Edge processing may filter, aggregate, route or enrich V2X messages before forwarding. The pipeline is designed so each operation has a clear purpose, bounded execution time and defined effect on message age. Expensive diagnostics, statistics or management transforms are decoupled from the primary path unless their worst-case timing is included. The product records enough stage-level telemetry to locate abnormal latency without logging sensitive payload content unnecessarily.

When processing rules require infrastructure context, that context has its own freshness and authority contract. A missing map, signal-state or policy object cannot be replaced by an old cached object beyond its approved lifetime merely to preserve output continuity. The gateway prefers explicit degraded output or suppression to silent extrapolation.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-MSG-004	Each processing stage shall have bounded execution and queueing behavior included in the released latency budget.	Stage-level profiling under worst-case traffic.
V2X-MSG-005	Cached context used to transform messages shall have a released validity lifetime and invalidation rule.	Cache-expiry and stale-context injection.
V2X-MSG-006	Diagnostic instrumentation shall preserve timing observability without materially changing the measured processing path.	Instrumented vs baseline timing correlation.

Verification, margin and future-change rule

Rule-engine, filtering algorithm, context source or logging changes reopen the affected message-validity and latency evidence. Optimization may reduce average latency only if worst-case validity and resource behavior remain bounded.

FAIL-SAFE DEGRADATION WHEN TIME SYNCHRONIZATION IS LOST

The product safety position is fail-safe degradation if C-V2X timing synchronization is lost. Degradation is defined at the service level: functions that require trustworthy absolute time or message age suppress output, mark the message as unqualified, or use an approved reduced function only while its prerequisites remain valid. The gateway does not present a free-running local clock as synchronized simply to maintain throughput.

The reaction is observable to management and downstream consumers, includes a reason and current time-quality state, and is reversible only after reacquisition criteria are met. Recovery avoids bursts of queued stale data. Functions independent of synchronized time may continue, provided their own data and authority conditions are satisfied and they cannot be mistaken for the degraded time-dependent service.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-SAFE-001	Loss of qualified synchronization shall transition each time-dependent service to its released degraded state within the allocated response.	Time-loss fault injection with service-state trace.
V2X-SAFE-002	Degraded operation shall be explicitly indicated and shall not label an unqualified local clock as synchronized.	Interface/status conformance test.
V2X-SAFE-003	Recovery from time loss shall discard or re-age queued data before normal forwarding resumes.	Reacquisition and stale-queue test.

Verification, margin and future-change rule

Time-source, holdover or service-allocation changes require renewed failure-response and recovery testing. The degraded-state definition is an interface contract and cannot be changed through calibration without impact review.

EXTERNAL DATA VALIDITY, PLAUSIBILITY & STALE-DATA CONTROL

V2X usefulness depends on external data that can be wrong even when communications remain healthy. The gateway therefore separates transport integrity from application plausibility. Vehicle-originated position, speed, heading or status fields and infrastructure-originated phase/state data are checked according to the supported profile and project rules. Plausibility checks do not invent physical truth; they identify clearly inconsistent, impossible, stale or internally contradictory inputs for rejection or degraded handling.

Data freshness uses qualified timestamps where required, while sequence or reception-age mechanisms may support profiles that do not carry absolute time. Duplicate and repeated messages are distinguished from valid high-rate updates. Invalid external data cannot increase command or forwarding permissiveness, and the gateway records rejection reasons suitable for engineering diagnostics without exposing credentials or private identifiers beyond authorized use.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-MSG-007	Transport-valid but stale or implausible data shall be treated according to the released message policy rather than assumed trustworthy.	Plausibility/freshness fault-injection matrix.
V2X-MSG-008	Duplicate, repeated and out-of-order messages shall be distinguished using profile-appropriate state.	Sequence/order negative testing.
V2X-MSG-009	Rejected data shall produce diagnostic reason information without widening disclosure of sensitive identifiers.	Diagnostic-content and privacy review.

Verification, margin and future-change rule

New data providers, field semantics or plausibility algorithms require message-specific regression with boundary values and realistic timing. A broadened acceptance window is treated as a functional change, not a benign calibration tweak.

WATCHDOG, HEALTH SUPERVISION & CONTROLLED RECOVERY

Supervision covers processor health, application heartbeat, radio readiness, time quality, network link state, storage condition, thermal state and credential-service availability. A single global heartbeat is not sufficient because one responsive process can coexist with a stalled message path. Health states are composed into service availability so external systems can see which capabilities are healthy, degraded or unavailable.

Recovery is staged from local service restart to subsystem reset and full reboot according to fault classification. Persistent reboot loops are bounded and reported rather than allowed to generate repeated network attachment or radio-transmit churn. First-fault information, reset reason and essential timing context are retained in a protected diagnostic record whose storage cannot exhaust the operational partition.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-DIAG-001	Health supervision shall cover independent indicators for compute, radio, time, network, storage and critical application paths.	Fault-injection coverage test.
V2X-DIAG-002	Recovery escalation shall be bounded and shall not create an uncontrolled reboot or reconnect loop.	Persistent-fault recovery test.
V2X-DIAG-003	First-fault and reset-reason data shall be preserved without consuming operational storage needed for message services.	Reset/log-retention stress test.

Verification, margin and future-change rule

Watchdog IC, supervisor, restart policy, persistent-log format or health-state composition changes require fault-injection regression for the affected coverage and recovery path.

SCMS PKI, HARDWARE TRUST ANCHOR & COMMAND AUTHORITY

The cybersecurity architecture uses an SCMS PKI with trust material protected by a hardware trust anchor. The gateway distinguishes identity and credential services from application message authority: possession of a valid credential does not allow a management channel to bypass message freshness, timing or local resource limits. Trust-anchor initialization, key use, certificate state and failure modes are explicitly defined for manufacturing, field operation and service.

Private key material is not exported in plaintext through normal diagnostic paths. Cryptographic operations are bounded so certificate verification or signing load cannot starve the low-latency message pipeline; resource budgeting includes realistic credential activity. Loss or invalidity of a security service rejects untrusted operations but does not disable local fault containment or time-quality degradation logic.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-SEC-001	SCMS trust anchors, credential roles and cryptographic-use authority shall be configuration controlled for each released variant.	Security architecture and provisioning audit.
V2X-SEC-002	Private keys shall remain protected by the released hardware-root-of-trust mechanism during normal use and service.	Key-handling negative and access-control test.
V2X-SEC-003	Security-service overload or failure shall not bypass local validity, timing or fault-containment behavior.	Concurrent crypto-load and fault test.

Verification, margin and future-change rule

Trust-anchor device, cryptographic provider, certificate role model or SCMS backend integration changes reopen provisioning, performance, recovery and negative-authorization evidence. Algorithm or key-policy changes are treated as cybersecurity-impacting release changes.

SECURE BOOT, SOFTWARE UPDATE, ROLLBACK & RECOVERY

Boot integrity establishes a trusted platform before radio and application services are declared operational. Boot chain, signed software authority, protected configuration, anti-rollback policy and recovery image are configuration controlled. An update is not complete merely because a new image boots; compatibility with hardware, radio firmware, containers, credentials and persistent data must also be verified.

Interrupted update, invalid signature, corrupted image or incompatible configuration results in a bounded recovery path. Rollback cannot restore a software version whose security or compatibility policy has been revoked. Update bandwidth and CPU load are isolated from active low-latency services or scheduled so the released service behavior is maintained.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-SEC-004	Only authenticated approved boot and application images shall execute in the normal operational state.	Modified-image and signature negative test.
V2X-SEC-005	Interrupted or invalid update shall recover to an approved compatible state without corrupting credentials or configuration.	Power-loss/update fault injection.
V2X-SEC-006	Rollback shall obey released security and compatibility policy and shall not reactivate a revoked image.	Rollback-policy and version-compatibility test.

Verification, margin and future-change rule

Bootloader, signing hierarchy, update transport, persistent-data schema or recovery strategy changes require end-to-end update regression, including power interruption, storage wear and concurrent-service behavior.

CREDENTIAL LIFECYCLE, ROTATION, REVOCATION & PRIVACY BOUNDARIES

SCMS credentials have lifecycle states that include provisioning, activation, rotation or replenishment, expiry, revocation and service replacement. The gateway maintains clear separation between device-management identity, infrastructure identity and V2X credential roles defined by the deployment. Credential renewal failure cannot be hidden by indefinite use of expired material, and time validity of certificates depends on qualified time where applicable.

Logs and remote telemetry minimize disclosure of credential identifiers while preserving enough information for authorized fault isolation. Field service procedures define who may initiate provisioning or recovery and under what physical or logical conditions. Replacement hardware receives new controlled identity material rather than copying private keys from the failed unit.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-SEC-007	Credential state transitions shall define authorization, time prerequisites, failure behavior and audit evidence.	Credential lifecycle state-machine test.
V2X-SEC-008	Expired or revoked credentials shall not be accepted solely to preserve service continuity.	Expiry/revocation negative test.
V2X-SEC-009	Replacement units shall use controlled re-provisioning rather than uncontrolled private-key cloning.	Service/provisioning process audit.

Verification, margin and future-change rule

Changes to certificate profile, renewal mechanism, privacy policy, logging identifiers or service roles reopen credential-state and privacy review. Backend availability assumptions remain explicit installation or service dependencies.

PASSIVE COOLING, THERMAL MARGIN & RADIO DERATING

The gateway is passively cooled, so enclosure conduction, internal airflow, component placement, solar or cabinet heating, RF transmit duty, processor load and ambient conditions determine junction temperature. The thermal model identifies hotspots in compute, RF power devices, power conversion and storage, and it is correlated to representative hardware. Thermal protection does not rely on average ambient temperature when short-duration solar or sealed-enclosure conditions can create higher internal temperatures.

If the design uses performance derating at high temperature, the effect on RF coverage, message throughput and latency is bounded and externally visible where necessary. Local hardware protection remains independent of high-level optimization. Temperature sensing faults use conservative behavior rather than allowing unbounded transmit power or compute load.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-THM-001	Thermal analysis shall cover released ambient, installation, RF duty and compute-load combinations with component tolerances.	Thermal model correlated to chamber/hotspot mapping.
V2X-THM-002	Any thermal derating shall have bounded impact on radio power, processing capacity and service availability.	Hot-condition functional and latency test.
V2X-THM-003	Invalid temperature data shall not increase RF or compute permissiveness beyond the released thermal envelope.	Sensor-fault and overtemperature test.

Verification, margin and future-change rule

Enclosure material, heatsink or interface, processor, radio PA, regulator or installation orientation changes reopen hotspot and passive-cooling correlation. End-of-life interface degradation is included where it can materially consume margin.

NEMA 4X OUTDOOR ENCLOSURE, SEALING & SERVICEABILITY

The roadside enclosure is NEMA 4X rated and must protect electronics while preserving RF feedthrough integrity, pressure equalization strategy, thermal conduction and field-service access. Seal design considers rain, wind-driven water, washdown exposure consistent with the selected installation, dust, UV, temperature cycling, pollutants and connector interfaces. The rating is verified on the released mechanical stack rather than inferred from component-level gasket data.

Opening the enclosure for service changes the environmental boundary and therefore requires controlled resealing, inspection and post-service checks. Drainage, condensation control and materials are selected so trapped moisture cannot degrade RF connectors, grounding, PCB insulation or corrosion-sensitive surfaces over life. Any vent, if used, is specified by function and qualification rather than a mandatory brand.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-MECH-001	The released enclosure stack shall demonstrate NEMA 4X performance with production-representative seals, connectors and antenna feedthroughs.	Ingress/environment qualification on representative assemblies.
V2X-MECH-002	Field-service procedures shall restore the released sealing and grounding condition after enclosure access.	Service-process inspection and post-service ingress check.
V2X-MECH-003	Condensation or trapped moisture shall not compromise RF, power, grounding or insulation functions over the qualified lifecycle.	Humidity/temperature-cycle and corrosion inspection.

Verification, margin and future-change rule

Gasket, coating, vent, connector, enclosure casting, fastener or service-process changes require targeted ingress, corrosion, thermal and RF-feedthrough impact assessment.

PART II - RF LINK BUDGET, COVERAGE & RECEIVER MARGIN

Detailed engineering assurance begins with a deployment-specific link budget rather than a universal range claim. Transmit power, feeder and connector loss, antenna gain and pattern, polarization, installation height, receiver sensitivity, interference allowance and environmental aging are kept as controlled inputs. Coverage predictions are correlated to representative installations because roadside clutter, vehicle blockage and local spectrum occupancy can dominate nominal free-space margin.

Receiver margin is assessed together with the selected message profile and required service availability. The gateway does not claim communication quality from RSSI alone; packet error behavior, channel occupancy and antenna-path health are considered. Field telemetry supports trend monitoring but does not replace qualification evidence for a new installation or antenna variant.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-RF-007	Each deployment class shall have a controlled RF link budget with explicit antenna, feeder, interference and environmental assumptions.	Link-budget review plus representative radiated correlation.
V2X-RF-008	Coverage claims shall be tied to defined installation and traffic conditions rather than a universal distance.	Field/anechoic correlation report.
V2X-RF-009	RF health metrics shall distinguish antenna-path degradation from external interference where evidence allows.	Faulted-feedline and interference characterization.

Verification, margin and future-change rule

Changes to antenna placement, cable length, transmit power, receiver chain or deployment geometry reopen link-budget correlation. A software-only release cannot grandfather RF coverage if the physical assumptions changed.

RF INTERFERENCE, COEXISTENCE & SELF-DESENSE CONTROL

The gateway may operate near cellular, Wi-Fi, GNSS, switching power supplies and high-speed digital interfaces. Coexistence analysis identifies strong in-band and out-of-band aggressors, intermodulation risk, harmonics, local-oscillator leakage and digital-noise coupling that can raise the effective receiver noise floor. PCB partitioning, filtering, shielding, clock planning and antenna separation are verified in the assembled enclosure with representative cables.

Self-desense is evaluated under simultaneous compute, Ethernet and transmit activity rather than with subsystems tested independently. Software may schedule or coordinate radios when the architecture supports it, but such coordination does not replace hardware filtering needed for credible worst-case interferers. Coexistence failures produce diagnosable degraded performance rather than unbounded retries that increase congestion.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-EMC-001	The released product shall characterize receiver desense and spurious behavior with relevant internal and external aggressors active.	Coexistence chamber and conducted injection test.
V2X-EMC-002	Mitigation shall not rely solely on software scheduling where required hardware isolation or filtering is absent.	Architecture review plus simultaneous-operation test.
V2X-EMC-003	Interference-induced reception loss shall use bounded retry or degradation behavior and shall not create uncontrolled network load.	Interference/flood recovery test.

Verification, margin and future-change rule

Clock source, PCB stack, high-speed interface, RF filter, shield or antenna-separation changes reopen coexistence measurements focused on the affected coupling paths.

GNSS JAMMING / SPOOFING INDICATORS & HOLDOVER BEHAVIOR

Time-source assurance includes faults beyond simple loss of antenna. The design observes receiver status and available consistency indicators so a clearly implausible or conflicting GNSS solution does not remain trusted merely because satellites are reported. Project-approved spoofing or jamming detection capability is treated as bounded evidence, not as a guarantee against every attack. The gateway combines source quality with oscillator holdover and service-level requirements.

When trust in GNSS is lost, the device records the reason, transitions to holdover if qualified and increases uncertainty with elapsed time. Once uncertainty exceeds the service limit, affected functions degrade. Reacquisition requires stability criteria and avoids an immediate jump to full trust from a single sample.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-TIM-007	GNSS quality evaluation shall distinguish loss, clearly invalid or conflicting data and qualified reception states supported by the receiver.	GNSS fault-scenario test.
V2X-TIM-008	Holdover uncertainty shall increase according to released oscillator and time-service evidence and shall gate time-dependent services.	Timed holdover drift characterization.
V2X-TIM-009	Reacquisition shall require released stability criteria before full synchronized status is restored.	Spoof/loss/recovery transition test.

Verification, margin and future-change rule

GNSS firmware, oscillator, antenna, quality algorithm or holdover model changes reopen fault-detection and uncertainty correlation. New detection features must not be credited beyond demonstrated coverage.

TIMESTAMP ACCURACY, CLOCK DRIFT & CROSS-DOMAIN CORRELATION

Radio, kernel, application and network timestamps can originate in different clock domains. The release design documents how those clocks are synchronized or correlated, what uncertainty is introduced at each crossing and how timestamp wrap, time-step events and reset are handled. A high-resolution counter is useful only if its relationship to the qualified reference time is known for the message contract.

Correlation testing uses instrumented events visible at multiple boundaries so software timestamps can be checked against external measurement. The product retains enough reserve for oscillator tolerance, scheduling jitter and calibration error over temperature. If the system changes reference time or performs a step correction, queued message age is recomputed or invalidated rather than preserved from an obsolete epoch.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-TIM-010	Clock-domain relationships and timestamp uncertainty shall be documented for all boundaries used in latency or message-age claims.	Clock architecture and uncertainty analysis.
V2X-TIM-011	Software timestamps shall be correlated to representative external events with calibrated instrumentation.	Cross-domain timing correlation test.
V2X-TIM-012	Clock step, wrap or reset shall not leave queued data with silently incorrect age.	Time-step/wrap/reset fault injection.

Verification, margin and future-change rule

Driver timestamping mode, clock tree, oscillator, kernel timekeeping or radio firmware changes trigger cross-domain correlation only where they affect the claimed timing path.

WCET, SCHEDULING JITTER & LATENCY UNDER PEAK LOAD

Worst-case latency depends on more than average CPU utilization. The product measures execution time, scheduling delay, interrupt load, driver backlog, network queueing and container contention under a defined peak workload that includes security, diagnostics and management activity. CPU frequency scaling and thermal throttling states are included if they are enabled in production. The test configuration is traceable so later software changes can be compared to the same boundary.

Latency reserves are not consumed silently by new features. A release gate identifies whether the budget remains positive at the maximum supported traffic and message mix. If the compute platform enters a degraded thermal or resource state, service capability is reduced in a bounded way rather than allowing queue growth to create arbitrarily stale outputs.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-TIM-013	Measured execution and scheduling latency shall retain an approved reserve under the maximum defined production workload.	WCET/jitter trace with concurrent-service stress.
V2X-TIM-014	Frequency scaling, throttling and interrupt load shall be included in timing verification when enabled in the released configuration.	Corner-state timing sweep.
V2X-TIM-015	Resource degradation shall bound queue depth or service rate so message age cannot grow without limit.	Overload and thermal-throttle test.

Verification, margin and future-change rule

Compiler, scheduler, CPU policy, interrupt topology, container quota or major application algorithm changes reopen only the timing paths whose WCET or queue behavior can change.

MESSAGE FRESHNESS, EXPIRY & REPLAY CONTROL

Freshness is enforced using the mechanisms available to each message class, such as qualified timestamp, sequence state, reception age or credential-associated replay protection. The gateway keeps transport receipt time separate from producer time so it can distinguish network delay from stale generation. Expiry windows are controlled parameters tied to the use case and may not be widened by a generic fallback when synchronization is uncertain.

Replay or duplicate detection state survives normal container restart only when doing so is safe and well defined; otherwise the service re-enters through a conservative reinitialization state. Cached data used for SPaT or other infrastructure functions is invalidated when its source authority changes or its lifetime expires. Logs capture replay rejection counts without retaining sensitive payloads longer than necessary.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-MSG-010	Each message class shall define the freshness and expiry mechanism used to determine whether data is current.	Profile review and age-boundary testing.
V2X-MSG-011	Replay, duplicate or stale data shall not be forwarded as a new current event when detection prerequisites are available.	Replay/duplicate injection.
V2X-MSG-012	Loss of synchronization shall not automatically widen expiry windows for time-dependent data.	Time-quality/freshness combined test.

Verification, margin and future-change rule

Freshness-rule, sequence-state, cache or credential-replay changes require message-class-specific regression. A larger timeout is treated as a functional risk change and must be justified, not hidden in calibration.

CONGESTION, QOS, RATE CONTROL & QUEUE BOUNDING

Roadside gateways can encounter bursts of V2X messages, backhaul congestion and management activity at the same time. Traffic classes, queue limits and scheduling policy protect time-sensitive messages while preventing low-priority flows from starving management needed for recovery. Rate control is bounded and observable; the product does not preserve throughput by retaining stale packets in deep queues.

Queue overflow has a defined disposition by traffic class, with drop or degradation behavior chosen to preserve freshness and system stability. Telemetry records occupancy, drops and overload duration so field investigations can distinguish RF loss from local congestion. Security filtering remains active during overload and is included in performance testing.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-NET-004	Each network or message traffic class shall have a released queue, priority and overload disposition.	QoS configuration review plus burst testing.
V2X-NET-005	Queueing shall be bounded so overload cannot convert current traffic into arbitrarily stale outputs.	Sustained-congestion test.
V2X-NET-006	Security and validity checks shall remain effective under the maximum supported traffic load.	Concurrent overload and negative-message test.

Verification, margin and future-change rule

Network driver, queue depth, scheduler, traffic classification or backhaul topology changes reopen overload and latency tests for the affected classes. Increasing buffers is not accepted as a timing fix without freshness analysis.

PACKET LOSS, DUPLICATION & OUT-OF-ORDER HANDLING

Wireless and packet networks can lose, duplicate or reorder messages without indicating a hardware fault. The gateway therefore defines how each supported message class handles missing updates, duplicate delivery and out-of-order arrival. It does not interpolate safety-relevant state beyond the project-approved behavior. Where sequence information is unavailable, the product relies on the profile-defined age and plausibility rules rather than inventing hidden sequence semantics.

Recovery after a burst loss avoids releasing a backlog of old data simply because connectivity returned. Counters and diagnostics identify loss patterns while keeping normal processing bounded. Upstream and downstream systems receive a status that allows them to distinguish unavailable data from valid zero or unchanged values.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-MSG-013	Loss, duplicate and out-of-order handling shall be defined for each supported message profile.	Packet-fault matrix and protocol conformance test.
V2X-MSG-014	Connectivity recovery shall not emit an aged backlog as if it were current data.	Link-interruption and restoration test.
V2X-MSG-015	Unavailable data shall be distinguishable from valid zero, unchanged or default values at the output interface.	Interface semantic test.

Verification, margin and future-change rule

Protocol-stack, buffering, retransmission or message-profile changes require targeted packet-fault regression. A new retry mechanism must be analyzed for both latency and duplicate risk.

MESSAGE PLAUSIBILITY, MISBEHAVIOR INPUTS & SAFE DEGRADATION

The gateway may receive cryptographically valid messages that are physically implausible or inconsistent. Application-level plausibility checks are bounded by the data available to the gateway and are not presented as proof of truth. They can reject impossible values, contradictory fields, invalid timing relationships or inputs outside the supported profile. Where a deployment includes a separate misbehavior authority or analytics service, that external function does not bypass local validity rules.

A suspicious input does not trigger broad denial of service by default. The gateway applies the released message-level reaction, records suitable evidence and preserves independent valid traffic where possible. Rate limiting and source isolation are used only within authorized policy so an attacker cannot exploit plausibility handling to exhaust CPU or storage.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-MSG-016	Plausibility rules shall define the fields, bounds, temporal relationships and reaction used for each supported message class.	Boundary and contradictory-field test.
V2X-MSG-017	A rejected source or message shall not unnecessarily suppress independent valid traffic outside the allocated reaction.	Mixed-validity traffic test.
V2X-MSG-018	Misbehavior handling shall be resource bounded and shall not create an unbounded logging or analysis load.	Adversarial-rate stress test.

Verification, margin and future-change rule

Plausibility algorithm, external misbehavior service, reaction policy or data-field change reopens targeted validity and denial-of-service testing. Detection claims remain limited to demonstrated rules.

SCMS CERTIFICATE PROVISIONING, ROTATION & REVOCATION

Certificate operations are treated as a lifecycle subsystem with factory, field and service states. Manufacturing injects or establishes device trust material under controlled authority, while operational SCMS credentials are activated and maintained according to the deployed backend. Provisioning steps identify what data is unique per unit, what can be regenerated, and which artifacts must be retained for traceability without exposing private keys.

Rotation or replenishment is verified under normal and interrupted connectivity. Expiry and revocation are respected even when network access is temporarily unavailable, using the project-defined offline policy. Recovery from a failed credential update returns to an approved state and does not leave mixed or partially committed credential sets.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-SEC-010	Factory and field credential provisioning shall be role controlled, traceable and bound to the serialized hardware identity.	Provisioning-process and unit genealogy audit.
V2X-SEC-011	Interrupted rotation or replenishment shall recover without exposing private material or leaving an ambiguous active set.	Credential-update fault injection.
V2X-SEC-012	Revoked or expired credentials shall follow the released offline or online policy and shall not be silently extended.	Revocation and expiry scenario test.

Verification, margin and future-change rule

SCMS backend, certificate profile, provisioning station, secure element or service process changes reopen the relevant manufacturing, field and recovery evidence. Production equivalence requires new provisioning correlation, not only electrical compatibility.

HARDWARE TRUST ANCHOR, KEY STORAGE & TAMPER RESPONSE

Protected hardware stores or mediates access to trust anchors and private keys. The released design specifies device identity, boot trust, V2X credential use, debug state, key import or generation path and authorized service operations. Physical tamper resistance is credited only to the capability actually provided by the selected hardware and enclosure; the PRD does not claim an unspecified certification or attack-resistance level.

Debug and manufacturing interfaces are disabled or access controlled according to lifecycle state. Loss of the trust-anchor device or detected key-storage fault makes dependent secure operations unavailable while local non-security fault handling remains intact. Service replacement follows re-provisioning procedures rather than copying secrets from a failed unit.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-SEC-013	Trust-anchor lifecycle state, debug access and key-use permissions shall be defined for manufacturing, operation and service.	Lifecycle and debug configuration audit.
V2X-SEC-014	Key-storage failure shall prevent dependent trusted operations without disabling independent local fault handling.	Trust-anchor fault injection.
V2X-SEC-015	Private key extraction shall not be available through normal diagnostic or service interfaces.	Authorized and unauthorized access negative test.

Verification, margin and future-change rule

Trust-anchor device, debug strategy, board layout or enclosure-access change triggers focused security, provisioning and service review. Claims about tamper resistance remain tied to supplier-guaranteed and verified capability.

NETWORK SEGMENTATION, MAINTENANCE ACCESS & SECURITY INDEPENDENCE

The gateway separates V2X message handling, local infrastructure networking and maintenance or management authority according to the deployment. A maintenance session may read diagnostics or perform approved configuration and update operations, but it cannot directly bypass message freshness, time quality, RF power limits or local fault containment. Network segmentation is supported by filtering, interface policy and least-privilege services rather than relying on naming conventions alone.

Loss of remote authentication or backend reachability fails closed for privileged operations while keeping local health and fail-safe degradation available. Management floods and malformed requests are included in resource tests because security control is ineffective if it consumes all compute needed for critical forwarding.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-SEC-016	Maintenance and management paths shall have explicit roles and shall not possess direct bypass authority over local validity or protection rules.	Role and authorization architecture review.
V2X-SEC-017	Loss of remote trust services shall reject privileged actions without disabling local safe degradation and diagnostics.	Backend-loss and negative-authorization test.
V2X-SEC-018	Management traffic shall be rate and resource bounded so security processing cannot starve critical V2X services.	Authenticated and unauthenticated flood test.

Verification, margin and future-change rule

Firewall policy, port exposure, maintenance role, remote-management protocol or network-zone topology changes require renewed negative-authorization, flood and service-independence verification.

POE SURGE, LIGHTNING, GROUNDING & COMMON-MODE STRESS

Outdoor Ethernet and power wiring can couple surge and common-mode stress into the gateway. The installation defines whether protection is internal, external or coordinated across both. Magnetics, surge devices, protective bonding, shield termination and PCB spacing are selected from the actual residual waveform and grounding architecture rather than from a generic outdoor label. Surge protection must not create an RF or Ethernet ground path that defeats the intended enclosure and antenna bonding concept.

Tests monitor not only survival but also unintended reset, corrupted configuration, time loss, false radio state and post-stress Ethernet or RF performance. Repetitive events and component aging are considered where they can reduce clamping margin. Field wiring instructions make the grounding dependency explicit.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-PWR-004	PoE and Ethernet surge protection and grounding shall be released with the installation residual-stress assumptions.	Protection coordination analysis and surge test.
V2X-PWR-005	Surge exposure shall not leave hidden corruption of configuration, timing, network or RF calibration.	Post-surge functional and data-integrity test.
V2X-PWR-006	Ground and shield termination shall be defined so protection and EMC behavior remain repeatable in the field installation.	Bonding inspection plus common-mode test.

Verification, margin and future-change rule

Ethernet magnetics, surge devices, shield termination, enclosure bonding or external protection changes reopen surge, EFT or common-mode and post-stress functional verification for the affected interface.

12 V / 24 V DC TRANSIENTS, HOLD-UP & RESET INTEGRITY

DC-fed variants are validated against the program-defined low-voltage source and transient profile. Input filtering and energy storage are sized so short interruptions, dips or feeder disturbances produce either continued bounded operation or a clean reset with no ambiguous radio or credential state. The design records reset cause and validates persistent configuration before restarting services.

Hold-up is treated as a system behavior rather than a capacitor value. It includes regulator UVLO, processor brownout, storage-write behavior and service shutdown sequencing. A repeated marginal supply shall not cause rapid radio reconnect loops or uncontrolled filesystem wear. Where source transients are installation specific, the unqualified values remain TBC and are closed in the deployment DVP.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-PWR-007	DC variants shall define released transient, interruption and source-impedance conditions for clean operation or reset.	Input profile sweep with reset-state capture.
V2X-PWR-008	Power loss during persistent writes shall recover without ambiguous configuration or credential state.	Power-cut during write or update test.
V2X-PWR-009	Repeated marginal input shall use bounded restart behavior and shall not create uncontrolled reconnect or storage-wear loops.	Repeated-dip endurance test.

Verification, margin and future-change rule

Regulator, bulk storage, filesystem, reset supervisor or feeder-assumption changes trigger targeted hold-up, brownout, endurance and conducted-EMI regression.

OUTDOOR THERMAL, HOT-SOAK, COLD-START & SOLAR-LOAD DVP

Outdoor qualification covers the installation-specific thermal environment, including cold soak, hot ambient, solar or cabinet heating where applicable, self-heating from RF transmit duty and compute load, and passive-cooling limits. Startup at thermal extremes is verified for oscillator acquisition, GNSS or time qualification, radio calibration, storage behavior and network attachment; simple survival without correct service timing is insufficient.

The test matrix includes steady hot operation, transition from cold to high load and recovery from thermally reduced service. Temperature gradients and enclosure surface temperature are measured alongside internal hotspots. Any unqualified ambient category remains a controlled program parameter rather than being inferred from unrelated electronics ratings.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-ENV-001	The released outdoor thermal envelope shall include passive-cooling assumptions, self-heating and installation solar or cabinet effects where applicable.	Thermal qualification plan and chamber correlation.
V2X-ENV-002	Cold and hot startup shall verify radio, time, storage and network readiness in addition to basic boot completion.	Thermal-extreme startup test.
V2X-ENV-003	Thermal service reduction and recovery shall preserve bounded latency and explicit availability status.	Hot-load derating and recovery test.

Verification, margin and future-change rule

Enclosure finish, heatsinking, installation orientation, processor or radio silicon or transmit-duty assumption changes require focused thermal delta analysis and regression.

CONDENSATION, SEALING, CORROSION & ENVIRONMENTAL ROBUSTNESS

NEMA 4X protection addresses water ingress, but long-life roadside reliability also depends on condensation control, seal compression, connector materials, coating, pollutants and galvanic compatibility. Temperature and humidity cycles are sequenced with powered operation where appropriate to expose leakage or corrosion mechanisms that a dry ingress test can miss. Moisture inside RF connectors or feedthroughs is treated as both an environmental and communication-margin concern.

Inspection criteria include seal condition, corrosion, insulation resistance, connector contact resistance and RF or antenna-path behavior after stress. Field-service instructions define replacement of seals and contamination controls. A cosmetic pass does not close qualification if electrical or RF margin has shifted beyond the released acceptance limits.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-ENV-004	Environmental qualification shall assess condensation and corrosion mechanisms that can affect power, RF, grounding or insulation performance.	Humidity and temperature cycle with post-stress electrical and RF checks.
V2X-ENV-005	Seal and connector acceptance shall include production-representative compression, assembly and service conditions.	Mechanical stack and service-cycle validation.
V2X-ENV-006	Post-environment RF and electrical performance shall remain within released acceptance limits, not merely remain functional.	Before and after margin comparison.

Verification, margin and future-change rule

Seal material, coating, connector plating, enclosure alloy or field-cleaning process changes trigger targeted environmental and corrosion requalification based on the affected mechanism.

ANTENNA FEEDTHROUGH, CONNECTOR & CABLE AGING

Antenna feedthroughs and cables combine RF, environmental and service risks. The released assembly controls connector torque or retention, weather sealing, bend radius, strain relief, shielding, grounding and cable-loss allowance. Mechanical vibration, thermal cycling and repeated service can change contact resistance or RF insertion loss before a complete open circuit appears.

Lifecycle tests correlate electrical continuity, return loss or equivalent RF indicators and visual or mechanical condition. The gateway uses field RF-health telemetry to detect trends, but service thresholds are established from qualification and fleet evidence rather than arbitrary signal-strength changes. Cable substitutions require frequency-appropriate loss and environmental data.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-RF-010	Feedthrough and cable assemblies shall have released RF loss, sealing, retention and installation limits.	Assembly drawing and BOM review plus RF characterization.
V2X-RF-011	Mechanical and environmental aging shall not consume antenna-path margin beyond the released service limit.	Vibration, thermal and service-cycle RF correlation.
V2X-RF-012	Cable or connector second sources shall demonstrate equivalent RF and environmental behavior before production approval.	Supplier comparison and delta qualification.

Verification, margin and future-change rule

Connector family, cable construction, routing, strain relief or service torque changes reopen only the antenna-path RF, mechanical and environmental evidence they affect.

RF CALIBRATION & AUTOMATED ANTENNA TUNING AT END OF LINE

Manufacturing includes automated RF calibration and antenna tuning as defined by the product baseline. The EOL station uses traceable fixtures, calibrated reference paths and configuration-controlled limits. Calibration records are bound to the serialized hardware and software or firmware state so field behavior can be traced to the exact factory result. Software trim is not allowed to hide an assembly defect such as a wrong component, poor connector or damaged antenna path.

Golden-unit and fixture-correlation checks detect drift in the production tester. Failed tuning or calibration remains a production blocker until disposition is approved; the unit is not released in a runtime-degraded mode simply because some communication is possible. Rework requires repeat of all affected RF checks.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-MFG-001	EOL RF calibration and tuning shall use traceable fixtures, calibrated references and released acceptance limits.	Fixture MSA or correlation and EOL record audit.
V2X-MFG-002	Calibration data shall be bound to serialized hardware and the released RF and software configuration.	Genealogy and configuration trace check.
V2X-MFG-003	RF calibration failure shall block production release until approved rework and affected retest are complete.	EOL fail and rework process audit.

Verification, margin and future-change rule

Tester hardware, cable set, reference standard, calibration algorithm, RF component or antenna process changes require fixture correlation and targeted production validation before equivalence is claimed.

TIMING CALIBRATION, GNSS VALIDATION & EOL TIME-CORRELATION

EOL testing confirms that the manufactured unit can acquire and maintain the expected time references within the released production acceptance limits. The station verifies relevant oscillator behavior, GNSS path connectivity, timestamp function and correlation between radio or application timing and a traceable reference. The test is not a substitute for full environmental holdover qualification; it is a production screen tied to known failure modes.

Fixture latency and cable delays are characterized so manufacturing measurements do not embed unknown tester offsets. Units with timing results outside the released range are not normalized by an unrestricted software offset. Calibration, if used, has bounded trim limits and retains raw pre- and post-adjustment values for traceability.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-MFG-004	EOL timing validation shall use a traceable reference and characterized fixture delays.	Time-reference calibration and fixture audit.
V2X-MFG-005	Any oscillator or timing trim shall remain inside approved limits and retain raw and corrected results.	Calibration-range and record review.
V2X-MFG-006	Timing failure at EOL shall block release rather than being accepted through wider runtime freshness windows.	EOL disposition and negative test.

Verification, margin and future-change rule

Time-reference equipment, GNSS fixture, oscillator supplier, trim algorithm or timestamp implementation changes require production correlation and targeted timing regression.

SCMS CERTIFICATE INJECTION, SERIALIZATION & UNIT GENEALOGY

Production provisioning links each unit serial number, hardware revision, software image, trust-anchor state and certificate-provisioning result. Root or manufacturing authority material is handled only by approved stations and roles. The process prevents duplicate identities, uncontrolled retries and cross-unit credential mix-up. Provisioning logs contain enough evidence for audit without copying private key material into ordinary manufacturing databases.

A unit that fails provisioning is quarantined until the failure is resolved through an approved process. Board replacement or secure-element replacement changes identity state and requires the designated re-provisioning path. Production software tools are version controlled and validated against the released hardware variant.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-MFG-007	Unit serialization and trust or certificate provisioning shall be one-to-one, traceable and protected against duplicate or cross-unit assignment.	Manufacturing genealogy and duplicate-injection test.
V2X-MFG-008	Provisioning logs shall retain necessary audit evidence without exposing private key material.	Log-content and access-control audit.
V2X-MFG-009	Secure-element or board replacement shall trigger the approved identity re-provisioning workflow.	Service and manufacturing replacement test.

Verification, margin and future-change rule

Provisioning station, secure-element supplier, certificate backend, manufacturing authority or genealogy database changes require end-to-end correlation before production release.

REMOTE TELEMETRY, RF DEGRADATION TRENDING & 15-YEAR LIFECYCLE

The lifecycle baseline uses remote telemetry to monitor RF signal degradation over a 15-year product horizon. Trend data may include receive quality, transmit diagnostics, antenna or tuning indicators, temperature, reset history and error counters where permitted. The analysis separates environment and traffic effects from hardware aging before recommending service. A single weak-signal event is not automatically classified as component degradation.

Lifecycle telemetry is rate limited, privacy aware and does not compromise low-latency message processing. Service thresholds are derived from qualification, production distribution and field evidence, then configuration controlled. Obsolescence and second-source changes are evaluated against link margin, timekeeping, cybersecurity, power, thermal and environmental assumptions rather than simple form, fit and function replacement.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-LIFE-001	Lifecycle telemetry shall capture approved indicators sufficient to trend RF-path and platform degradation without exhausting operational resources.	Long-duration telemetry and resource test.
V2X-LIFE-002	Service thresholds shall be traceable to qualification, production or field evidence and shall not be inferred from a single metric alone.	Reliability and field-data review.
V2X-LIFE-003	15-year lifecycle support shall include controlled obsolescence and second-source impact review for critical RF, timing, power and security components.	Lifecycle and obsolescence plan audit.

Verification, margin and future-change rule

Changes to telemetry schema, service thresholds, retention policy or field analytics require validation against privacy, resource and diagnostic-effectiveness requirements. Long-life equivalence is never claimed solely from initial electrical similarity.

SYSTEM DVP - LATENCY, RF, TIMING & FAIL-SAFE DEGRADATION

The critical DVP combines radio, timing, compute and network stresses that can interact in service. It covers representative channel conditions, peak message load, concurrent security activity, GNSS acquisition, loss and holdover, backhaul congestion, container restart, thermal load and power disturbance. Objective pass or fail criteria are fixed before execution and identify the exact message profile, hardware and software configuration, antenna system and instrumentation.

The DVP retains raw timestamp traces, RF measurements, queue and resource telemetry and fault-state logs so a pass can be independently reviewed. Tests exercise both nominal sub-20 ms mission performance at the defined boundary and the correct degraded behavior when prerequisites are unavailable. A nominal average-latency result cannot compensate for an unsafe freshness or time-quality failure.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-DVP-001	System DVP shall cover combined RF, timing, compute, network, security and environmental conditions representative of release.	Configuration-controlled DVP matrix and raw records.
V2X-DVP-002	Latency pass or fail shall use the released boundary and include distribution or worst-case evidence, not average-only reporting.	Timestamp data analysis and review.
V2X-DVP-003	Time-loss and invalid-data tests shall verify the released degraded states as part of the same production-equivalent configuration.	Fault-injection DVP report.

Verification, margin and future-change rule

Incomplete critical DVP keeps the verification gate open. A later change may reuse evidence only after documented impact analysis shows the relevant stress, timing, RF and authority assumptions remain bounded.

ENVIRONMENTAL / EMC / OUTDOOR QUALIFICATION DVP

Environmental release evidence is sequenced so mechanical, ingress, thermal, humidity, surge and EMC stresses reveal cumulative effects that isolated tests can miss. The matrix is selected from the actual roadside installation and project obligations; this PRD does not invent a universal environmental class beyond NEMA 4X and the supplied deployment concept. Functional monitoring during susceptibility tests watches time quality, message validity, unintended reset, radio state, network behavior and configuration integrity.

Post-stress checks include RF calibration shift, antenna and feedline performance, Ethernet or PoE operation, GNSS or time behavior, enclosure sealing and diagnostic state. Qualification is performed on production-representative hardware with released firmware and critical mechanical stack. Missing environmental evidence blocks the affected hardware or installation variant from release; it is not represented as a runtime derate command.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-DVP-004	Environmental and EMC DVP shall be derived from the released roadside installation and shall monitor functional degradation behavior during susceptibility.	Qualification-plan review and test reports.
V2X-DVP-005	Post-stress acceptance shall include RF, timing, network, sealing and configuration-integrity checks where affected.	Sequential qualification comparison.
V2X-DVP-006	An unqualified environmental variant shall remain unreleased until the required evidence is closed.	Release-gate and qualification status audit.

Verification, margin and future-change rule

Installation relocation, enclosure change, antenna or feedthrough change or revised surge or EMC environment triggers targeted delta qualification. Prior reports are reused only inside documented similarity bounds.

SUPPLIER / SECOND-SOURCE CONTROL & RELEASE-EVIDENCE CLOSURE

Second-source approval is mechanism based. Critical substitutions such as RF amplifiers, filters, oscillators, antennas, Ethernet magnetics, regulators, storage, trust anchors or enclosure materials are reviewed for the parameters that affect the gateway rather than only nominal datasheet equivalence. Supplier process changes and firmware-bearing components receive the same scrutiny when they can alter RF, timing, security, thermal or lifecycle behavior.

Final release requires a controlled architecture and ICD, parameter or TBC closure, hardware and software identities, RF calibration state, SCMS provisioning evidence, DVP status, manufacturing and EOL capability, environmental qualification and lifecycle change-impact record. Open high-severity mechanisms or unresolved production equivalence prevent release. Any project waiver remains explicit and traceable to approved authority and residual risk.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
V2X-REL-001	Critical supplier or second-source changes shall be approved only after parameter-level similarity and affected regression evidence are closed.	Supplier delta analysis and qualification review.
V2X-REL-002	Release evidence shall uniquely identify hardware, software, RF calibration, credential provisioning state, mechanical variant and DVP configuration.	Final configuration and evidence-index audit.
V2X-REL-003	Open high-severity technical, qualification or production-equivalence gaps shall prevent final release of the affected variant.	Release board or gate evidence review.

Verification, margin and future-change rule

Future product variants may reuse evidence only when change-impact analysis proves that the changed item is outside none of the linked electrical, RF, timing, cybersecurity, environmental, manufacturing and lifecycle assumptions. Otherwise the affected analyses and tests are reopened before equivalence is claimed.